# Flow File Map

ApprovalFlow · Where each action in the pipeline actually lives — companion to the Flow Cheatsheet. Paths relative to repo root.

## Intake

| Action | File | Function / class |
|---|---|---|
| POST /api/invoices handler | `services/intake/approvalflow_intake/app.py` | `submit_invoice` |
| Pydantic schema validation | `services/shared/approvalflow_shared/contracts.py` | `InvoiceSubmission` |
| Correlation id resolution | `services/shared/approvalflow_shared/correlation.py` | `CorrelationIdMiddleware`, `get_correlation_id`, `new_correlation_id` |
| Business logic entry, tracking_id = correlation_id | `services/intake/approvalflow_intake/service.py` | `IntakeService.submit` |
| Idempotency-Key claim | `services/intake/approvalflow_intake/service.py` | `submit (step 1)` |
| Fingerprint property (vendor\|invoiceNumber\|total) | `services/shared/approvalflow_shared/contracts.py` | `InvoiceSubmission.fingerprint` |
| Fingerprint claim / duplicate branch | `services/intake/approvalflow_intake/service.py` | `submit (step 2)` |
| Atomic claim primitive (SETNX-style) | `services/intake/approvalflow_intake/dapr_adapters.py` | `DaprSubmissionStore._claim` |
| Persist SubmissionRecord (Redis write) | `services/intake/approvalflow_intake/service.py` | `submit (step 3)` |
| Publish item.submitted (RabbitMQ) | `services/intake/approvalflow_intake/service.py` | `submit (step 3)` |
| Publisher implementation | `services/intake/approvalflow_intake/dapr_adapters.py` | `DaprPublisher` |
| GET /api/invoices/{id} (status view) | `services/intake/approvalflow_intake/app.py` | `get_status` |
| Loop-back: consume item.status-changed | `services/intake/approvalflow_intake/app.py` | `on_status_changed` |
| Apply the status update | `services/intake/approvalflow_intake/service.py` | `apply_status_change` |

## Decision — policy engine, agent, router, workflow

| Action | File | Function / class |
|---|---|---|
| Consume item.submitted, schedule workflow | `services/decision/approvalflow_decision/service/app.py` | `on_item_submitted` |
| Idempotency check before scheduling | `services/decision/approvalflow_decision/service/app.py` | `on_item_submitted (get_workflow_state)` |
| Orchestration: decide → publish → (pause) → resume | `services/decision/approvalflow_decision/service/decision_core.py` | `decide` |
| Deterministic policy rules (policy.md §1–5) | `services/decision/approvalflow_decision/policy_engine.py` | `evaluate, _evaluate_category` |
| Fail-safe assessment (agent unreachable) | `services/decision/approvalflow_decision/service/decision_core.py` | `_fail_safe_assessment` |
| Agent orchestration (LLM call) | `services/decision/approvalflow_decision/agent/assessor.py` | `DecisionAgent.assess` |
| Agent tools (fetch_policy, lookup_vendor) | `services/decision/approvalflow_decision/agent/tools.py` | — |
| Provider selection (stub/ollama/groq/gemini/openai) | `services/decision/approvalflow_decision/agent/client.py` | `build_chat_client` |
| Offline stub agent (used by CI / verify.py) | `services/decision/approvalflow_decision/agent/stub_client.py` | `StubChatClient` |
| **The router — the whole M12/F10 proof** | `services/decision/approvalflow_decision/router.py` | `route` |
| route → status mapping | `services/decision/approvalflow_decision/models.py` | `ROUTE_TO_STATUS` |
| Dapr Workflow orchestrator | `services/decision/approvalflow_decision/service/workflow.py` | `decision_workflow` |
| Activity: run decide() | `services/decision/approvalflow_decision/service/workflow.py` | `decide_activity` |

| | | |
|---|---|---|
| Activity: persist + publish outcome | services/decision/approvalflow_decision/service/workflow.py | publish_outcome_activity |
| Activity: apply the human's decision on resume | services/decision/approvalflow_decision/service/workflow.py | apply_human_decision_activity |
| Fetch thresholds (with fail-safe fallback) | services/decision/approvalflow_decision/service/config_client.py | fetch_thresholds |
| GET /api/decisions/{id} (audit read) | services/decision/approvalflow_decision/service/app.py | get_decision |
| POST /api/decisions/{id}/human-decision (resume) | services/decision/approvalflow_decision/service/app.py | human_decision |

### Config

| Action | File | Function / class |
|---|---|---|
| GET/PUT /config/thresholds | services/config/approvalflow_config/app.py | build_app handlers |
| Hardcoded fallback thresholds | services/config/approvalflow_config/defaults.py | DEFAULT_THRESHOLDS |
| Threshold validation (bounds, hard cap) | services/config/approvalflow_config/models.py | Thresholds, CEILING_HARD_CAP |

### Payment — the saga

| Action | File | Function / class |
|---|---|---|
| Consume payment.requested, schedule workflow | services/payment/approvalflow_payment/app.py | on_payment_requested |
| Saga orchestrator (reserve→pay→confirm→complete) | services/payment/approvalflow_payment/workflow.py | payment_workflow |
| Reserve budget | services/payment/approvalflow_payment/workflow.py | reserve_activity / _reserve |
| Execute payment (idempotency-key-before-call) | services/payment/approvalflow_payment/workflow.py | pay_activity / _pay |
| Compensation: release budget | services/payment/approvalflow_payment/workflow.py | release_activity / _release |
| Confirm / complete / fail | services/payment/approvalflow_payment/workflow.py | confirm_activity, complete_activity, fail_activity |
| Budget ledger (ETag first-write concurrency) | services/payment/approvalflow_payment/budget.py | BudgetLedger |
| Budget state store adapter | services/payment/approvalflow_payment/budget_dapr.py | DaprBudgetStore |
| Fake payment gateway | services/payment/approvalflow_payment/provider.py | execute_payment |
| GET /api/budgets/{dept}, GET /api/payments/{id} | services/payment/approvalflow_payment/app.py | get_budget, get_payment |

### Approval — the escalation queue

| Action | File | Function / class |
|---|---|---|
| Consume escalation.requested, enqueue | services/approval/approvalflow_approval/app.py | on_escalation |
| GET /api/approvals (the queue) | services/approval/approvalflow_approval/app.py | list_queue |
| POST /api/approvals/{id}/decision — claim→lease→side effects→mark done | services/approval/approvalflow_approval/app.py | decide |
| Side effect: resume the workflow | services/approval/approvalflow_approval/app.py | DaprDecisionGateway.raise_human_decision |
| Side effect: publish escalation.resolved | services/approval/approvalflow_approval/app.py | DaprDecisionGateway.publish_resolved |
| Claim / lease / mark-done primitives | services/approval/approvalflow_approval/store.py | DaprQueueStore (try_resolve, try_begin_side_effects, mark_side_effects_completed) |

## Notification & Reporting (pure sinks)

| Action | File | Function / class |
|---|---|---|
| Consume item.submitted (record who to notify later) | `services/notification/approvalflow_notification/app.py` | `on_item_submitted` |
| Consume item.status-changed, deliver if terminal | `services/notification/approvalflow_notification/app.py` | `on_status_changed` |
| Terminal-status allowlist + inbox dedupe | `services/notification/approvalflow_notification/service.py` | `TERMINAL_STATUSES, InboxStore.append` |
| GET /api/notifications | `services/notification/approvalflow_notification/app.py` | `handler` |
| Consume all 6 tracked topics | `services/reporting/approvalflow_reporting/app.py` | `build_app subscriptions` |
| Metrics + audit recording (dedupe-in-same-write) | `services/reporting/approvalflow_reporting/service.py` | `record, update_metrics, append_audit` |
| GET /api/reports/summary, GET /api/audit/{id} | `services/reporting/approvalflow_reporting/app.py` | `handlers` |

## Gateway & shared kernel

| Action | File | Function / class |
|---|---|---|
| Routing table, rate limiting, correlation-id fallback | `gateway/nginx.conf` | `location blocks, limit_req_zone, map $corr_id` |
| Header forwarding to backends | `gateway/proxy_params.conf` | `proxy_set_header X-Correlation-Id` |
| All event/contract shapes (camelCase ↔ snake_case) | `services/shared/approvalflow_shared/contracts.py` | `_CamelModel + every event class` |
| Correlation id plumbing | `services/shared/approvalflow_shared/correlation.py` | `CorrelationIdMiddleware` |
| Dapr HTTP client (state/invoke/pub-sub) | `services/shared/approvalflow_shared/daprio.py` | `DaprIO` |
| Role-based auth guard | `services/shared/approvalflow_shared/auth.py` | `require_role` |
| Shared FastAPI app factory + logging setup | `services/shared/approvalflow_shared/web.py, logging.py` | `create_app, configure_logging` |